

Course No. : **EEE/ECE F311**Course Title: **Communication Systems****Max. Marks : 30****Max. Time : 1 hr.**

Name:

ID No.

Tut. Section:

Instructions: Please write down your final answers clearly and in a simplified form. Each question carries **2 marks**. Use the backside of this main answer book for rough work. Avoid overwriting, as overwritten answers will not be rechecked.

Q. 1. Let $g(t) = \sum_{m=-\infty}^{\infty} \delta(t - mT_0)$, a periodic delta function with the fundamental period T_0 . The Fourier transform of $g(t)$ denoted by $G(f)$ is $\sum_{n=-\infty}^{\infty} \frac{\delta(f - n\beta)}{\alpha}$. Here, $\alpha = \text{-----}$, $\beta = \text{-----}$.

Ans. $\alpha = T_0$; $\beta = \frac{1}{T_0}$.

Q. 2. Consider the figure of merit (FoM) of single-tone AM and wideband FM receivers. If the FoM of wideband FM is 72 times the FoM of AM with 100% modulation, the modulation index of FM is -----.

The ratio of the approximate bandwidth of the WBFM signal to the single-tone frequency f_m is -----.

Ans. $\beta = 4$; $2(\beta + 1) = 10$.

Q. 3. Suppose a Direct sequence spread spectrum system has an information bit rate of 1.8 Kbps and a chip rate of 360 Kbps. The chip duration (μs) is ----- . The processing gain (in dB) $\approx$ -----.

Ans. $2.8 \mu s$; 23 dB .

Q. 4. A communication system uses a binary FSK transmitter (Tx) and a coherent FSK receiver. The Tx transmits binary data at 10 Mbps using a sinusoidal carrier having amplitude $1 \mu V$. Assuming AWGN channel with two-sided PSD equal to 10^{-20} W/Hz , the ratio $\frac{E_b}{N_0} (\text{dB}) \approx \text{-----}$, and the average probability of bit error is approximately equal to -----.

Ans. $\frac{E_b}{N_0} (\text{dB}) \approx 6.99 \text{ dB}$; $BEP \approx 0.0127$ (or) 1.27×10^{-2} .

Q. 5. Consider the transmission of binary PAM data over a baseband channel. The transmission bandwidth requirement is 72.0 KHz . Suppose that the transmitter uses a raised cosine pulse shaping filter with roll-off factor α . Note that

the Nyquist bandwidth is equal to $\frac{1}{2T_b}$, where T_b denotes the bit duration. If the bit duration is 10 microseconds,

α will be equal to ----- . If the roll-off factor is decremented by 25% , the transmission bandwidth corresponding to the new α is $\approx$ -----KHz.

Ans. $\alpha = 0.44$; Bandwidth ≈ 66.5 KHz

Q. 6. Consider single tone PCM. A continuous-time signal has a voltage range of -2 V to 2 V. The signal is sampled uniformly at the Nyquist rate and uniformly quantized. If each quantization level is encoded to 8 bits per sample, the resulting signal has ----- quantization levels, and the step size is $\approx$ -----.

Ans. $L = 2^8 = 256$; Step size $= \frac{1}{64} \approx 0.0156$ V.

Q. 7. Four message signals, each band-limited to 2.5 KHz are sampled at Nyquist rate. The resulting PAM samples are pulse code modulated and transmitted over a single channel after TDM. The number of quantization levels used is 512. Two synchronization bits are used for framing. The bit rate and the bit duration of the system are -----and -----, respectively.

Ans. 190 Kbps ; 5.3 μ s.

Q. 8. Suppose that $R = 10\text{ K}\Omega$, $T = 290\text{ K}$, and the bandwidth of observation is 1 MHz. What is the approximate value of root mean square (RMS) noise voltage (in μ V)? -----.

The noisy resistor produces noise power $\approx$ -----femtowatt.

(Note: Boltzmann's constant $K = 1.38 \times 10^{-23}$ joules per degree Kelvin.)

Ans. 12.65 μ V ; 4.00 femtowatts.

Q. 9. A WSS random process $X(t)$, whose mean value is 1, is applied to the input of an LTI system with impulse response $h(t) = \exp(-\pi t^2)$, $t \in R$. The mean value of output $Y(t)$ of the system is equal to ----- . Let $X(t) = 1 + 2 \cos(2\pi f_0 t + \Theta)$, where $\Theta \sim U[0, 2\pi]$. The value of ACF $R_X(\tau)$ at $\tau = 0$ is -----.

Ans. $\mu_X H(0) = 1$; $R_X(0) = 3$.

Q. 10. What is the approximate channel capacity of the telephone channel when the received SNR is 30 dB

and the channel bandwidth is equal to 20 KHz? ----- (Assume AWGN channel.)

Let $R_b = C$. The approximate spectral efficiency is -----.

Ans. $C = 199.3 \text{ Kbps}$; 9.97 bps/Hz .

Q.11. The random variables Y_1 and Y_2 have the joint probability density function (pdf)

$$p_{Y_1 Y_2}(y_1, y_2) = \frac{k}{(b_1^2 + y_1^2)(b_2^2 + y_2^2)}, \quad b_1, b_2 > 0.$$

The constant $k =$ -----. The random variables Y_1 and Y_2 are -----.

Ans. $\frac{b_1 b_2}{\pi^2}$; Statistically independent.

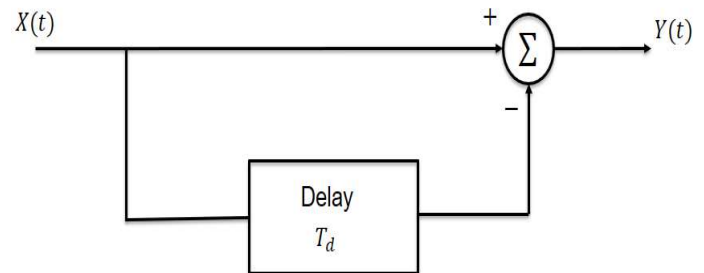
Q.12. Let $H(f)$ denote the transfer function of the Comb

Filter. The maximum value of $|H(f)|^2 =$ -----.

Suppose $T_d \sim U[-1, 1]$. Determine $E\{|H(f)|^2\}$ in terms of $\text{sinc}(\cdot)$ function.

Ans. $|H(f)|_{\max}^2 = 4$;

$$E\{|H(f)|^2\} = 2 (1 - \text{sinc}(2f)).$$



** END OF PART – A **

Course No. : **EEE/ECE F311**

Course Title: **Communication Systems**

Max. Marks : 50

Max. Time : 120 Min.

Name:

ID No.

Instructions: Use notation consistently. You may use standard results or formulae by stating them explicitly. Provide key intermediate steps. Highlight simplified final answers in rectangular boxes. Please avoid overwriting, as overwritten answers will not be rechecked.

Note: Full credit to be given only for complete solution (with key intermediate steps / computations).

Q. 1. [Gram-Schmidt Orthogonalization Procedure (GSOP)] Consider the four finite-energy, finite-duration signals shown in Fig. 1. Note: Do not change the order of the signal set.

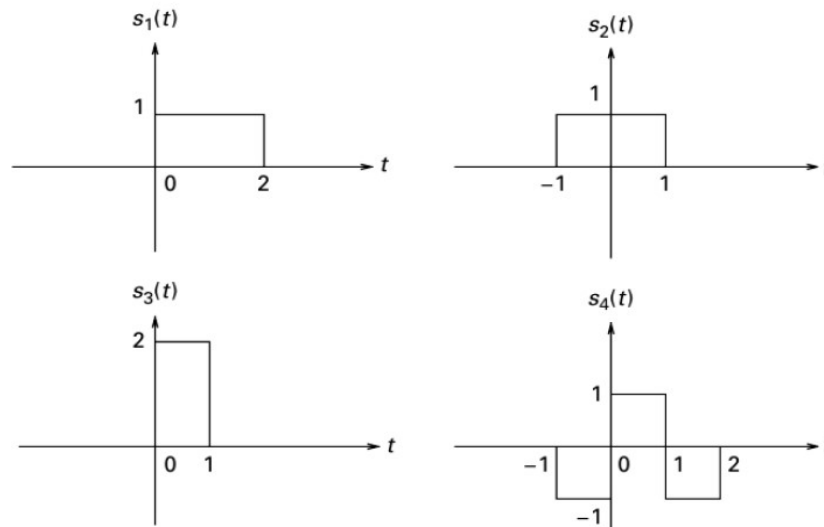


Fig. 1

(i) Apply the GSOP to find an orthonormal basis set. Express each signal in terms of basis functions.

(ii) Sketch orthonormal basis. Sketch signal space. Clearly label the basis and signal vectors.

(iii) Does the given signal set linearly independent (LI)? Justify?

[6 + 5 + 1.5 Marks]

Ans. (i) [1 + 2 + 2 + 1] Using the GSOP, show that the required number of orthonormal basis functions is three ($N = 3$), which are denoted by $\psi_1(t)$, $\psi_2(t)$, $\psi_3(t)$.

Signal space representation (vector form):

$$s_1 = [\sqrt{2}, 0, 0], \quad s_2 = \left[\frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{\sqrt{2}}, 0 \right]$$

$$s_1(t) = \sqrt{2}\psi_1(t),$$

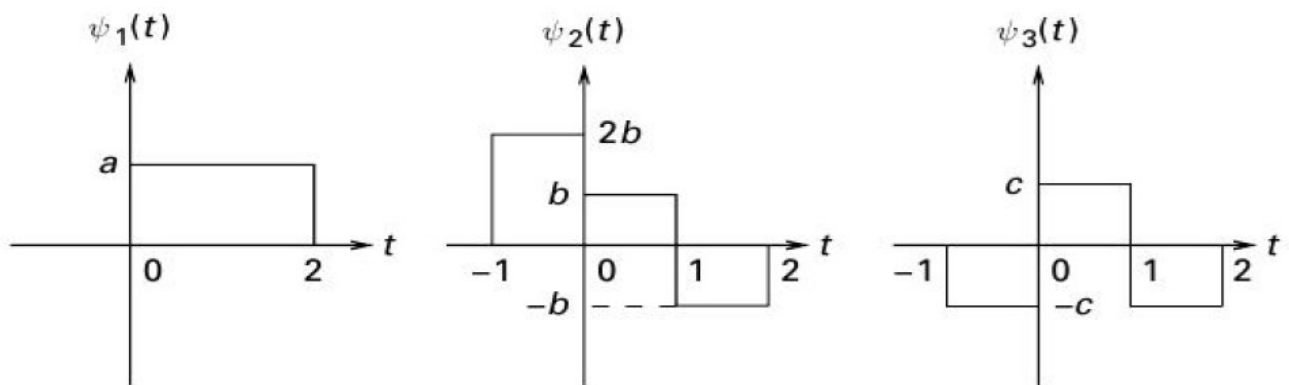
$$s_2(t) = \frac{1}{\sqrt{2}}\psi_1(t) + \frac{\sqrt{3}}{\sqrt{2}}\psi_2(t),$$

$$s_3(t) = \sqrt{2}\psi_1(t) + \frac{\sqrt{2}}{\sqrt{3}}\psi_2(t) + \frac{2}{\sqrt{3}}\psi_3(t),$$

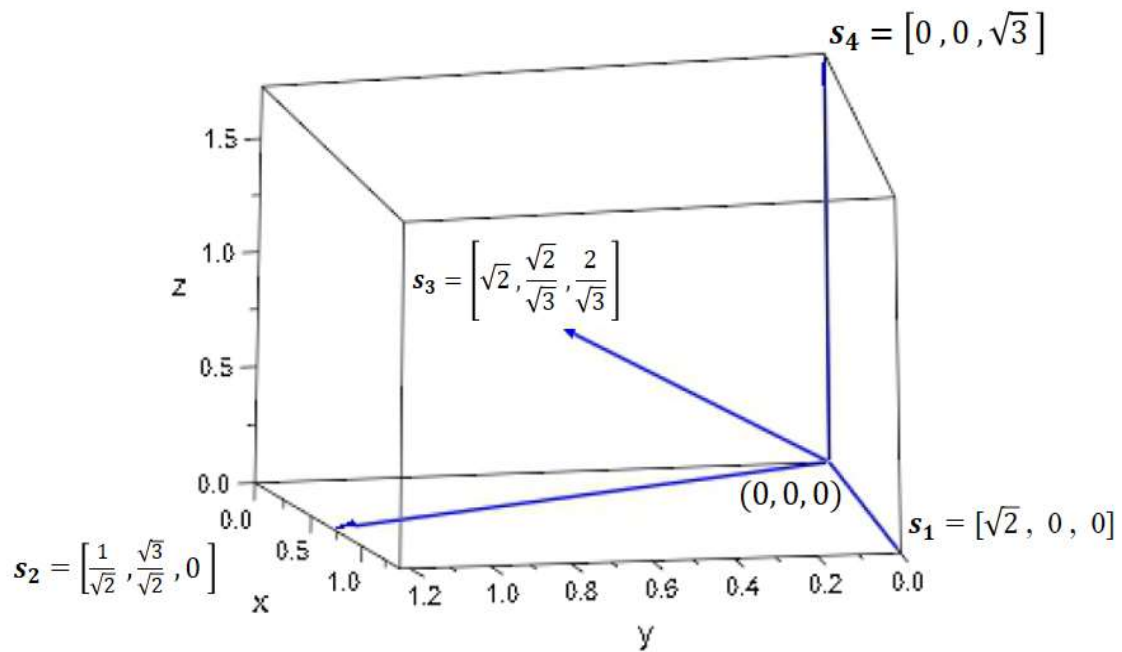
$$s_4(t) = \sqrt{3}\psi_3(t).$$

$$s_3 = \left[\sqrt{2}, \frac{\sqrt{2}}{\sqrt{3}}, \frac{2}{\sqrt{3}} \right] \quad s_4 = [0, 0, \sqrt{3}]$$

(ii) [3 + 2] Sketch the basis:



Here, $a = \frac{1}{\sqrt{2}}$, $b = \frac{1}{\sqrt{6}}$, $c = \frac{1}{\sqrt{3}}$.



(iii) Given number of signals $M = 4$.

Since $N < M$, given signals are linearly dependent. [1.5]

Q. 2. [4-PAM Constellation and SEP] [2 + 2.5 + 6 + 2 Marks]

Consider the constellation shown in Fig. 2. Consider transmission of symbols through AWGN with two-sided power spectral density $\frac{N_0}{2}$.

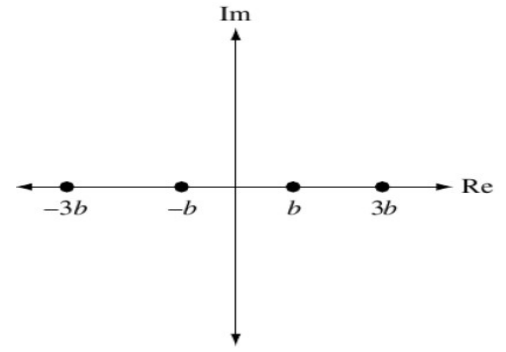
Assume that all symbols are equally likely.

(i) Let $b = \frac{1}{\sqrt{5}}$. Determine the average energy per symbol.

(ii) Neatly sketch the constellation and show the decision regions (labeling from left to right) for the four symbols.

(iii) Derive average symbol error probability (SEP) in terms of Q function. Express SEP in terms of $\text{erfc}(\cdot)$.

(iv) Derive the Chernoff bound for the SEP derived in (iii).



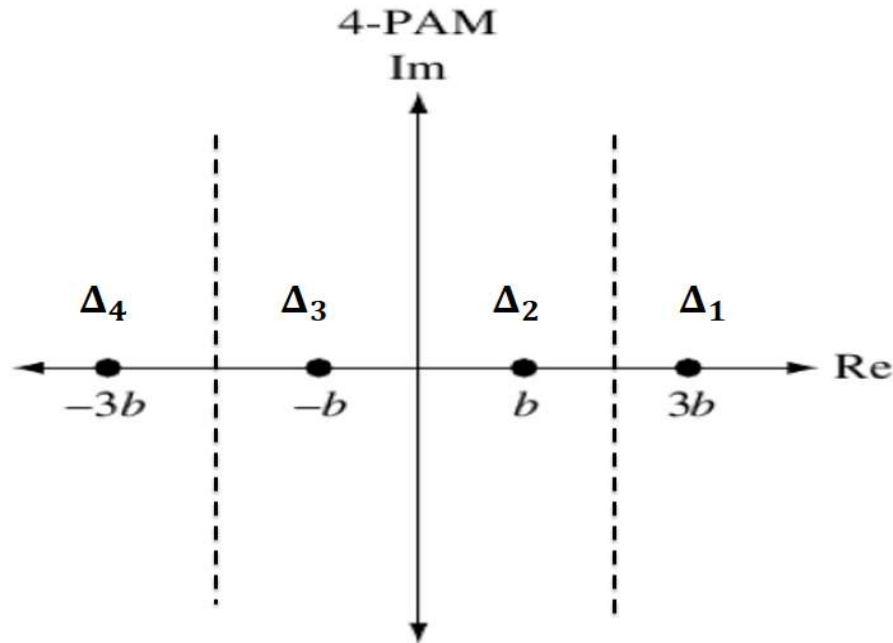
[P.T.O]

Fig. 2

Ans.

(i) Average energy per symbol = $5b^2 = 1$. [2]

(ii) Draw perpendicular bisectors for each segment joining two constellation points and identify the decision regions. Let the decision regions (from left to right) denoted by $\Delta_1, \Delta_2, \Delta_3, \Delta_4$. [2.5]



(iii) Consider the following scenario:

$x = -3b$ transmitted: Let y denote the received symbol.

$$\begin{aligned}\mathcal{P}(\mathcal{E}|x = -3b) &= 1 - \mathcal{P}(\mathcal{C}|x = -3b), \\ &= 1 - \mathcal{P}(y < -2b|x = -3b), \\ &= Q\left(\sqrt{\frac{2b^2}{N_0}}\right),\end{aligned}$$

where $\frac{N_0}{2}$ is the power spectral density of additive Gaussian noise, Note that, by symmetry, $\mathcal{P}(\mathcal{E}|x = -3b) = \mathcal{P}(\mathcal{E}|x = 3b)$.

Similarly, we can easily compute error probabilities for other symbols:

$$\begin{aligned}\mathcal{P}(\mathcal{E}|x = -b) &= \mathcal{P}(\mathcal{E}|x = b), \\ &= 2Q\left(\sqrt{\frac{2b^2}{N_0}}\right).\end{aligned}$$

Therefore, the average symbol error probability (P_e) is given by $\frac{3}{2}Q\left(\sqrt{\frac{2b^2}{N_0}}\right) = \frac{3}{2}Q\left(\sqrt{\frac{2}{5N_0}}\right) = \frac{3}{4}\text{erfc}\left(\sqrt{\frac{1}{5N_0}}\right)$.

$$\textbf{(Recall: } Q(y) = \frac{1}{2} \textit{erfc}\left(\frac{y}{\sqrt{2}}\right) \textbf{)} \quad [2 + 2 + 2]$$

(iv) Chernoff bound:

$$\textbf{Recall: } Q(t) \leq e^{-\frac{t^2}{2}}, \quad t > 0.$$

$$P_e = \frac{3}{2} Q\left(\sqrt{\frac{2}{5N_0}}\right) \leq \frac{3}{2} e^{-\frac{1}{5N_0}} \quad [2]$$

Q. 3. [Quantizer Model's Entropy] Consider the input-output characteristic of the uniform quantizer shown in Fig. 3. Assume that input X is modeled as a Gaussian random variable (GRV) of zero mean ($\mu = 0$) and unit variance ($\sigma^2 = 1$). Recall that Q –function is defined as

$$Q(t) = P(X > t) = \int_t^{\infty} p(x) dx,$$

where $p(x)$ denotes the probability density function of GRV X .

Let x_j denote the representation level of the quantizer. Denote the four quantization levels (bottom quantization level to top quantization level) as x_1, x_2, x_3 , and x_4 .

(i) Compute the following probabilities:

$$\begin{aligned} P(x_1) &= P\{-\infty < X < -1\}, & P(x_2) &= P\{-1 < X < 0\}, \\ P(x_3) &= P\{0 < X < 1\}, & P(x_4) &= P\{1 < X < \infty\}. \end{aligned}$$

How these probabilities are related? Comment.

(Note: Use the Q –function table.) [6.5 Marks]

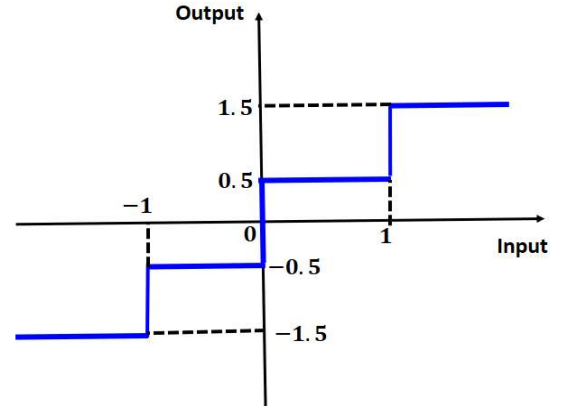


Fig. 3

(ii) Write down the expression for the entropy in terms of the probabilities $P(x_j)$, $j = 1, 2, 3, 4$. Compute the entropy. [6 Marks]

(i) Verify that $P(x_1) = P\{-\infty < X < -1\} = Q(1) = 0.1587 = P(x_4)$.

Further, verify that $P(x_2) = P\{-1 < X < 0\} = Q(-1) - Q(0) = 0.3413 = P(x_3)$.

Observe that $P(x_1) = P(x_4) = p_1$ and $P(x_2) = P(x_3) = p_2$.

[3 + 3 + 0.5]

(ii) The expression for entropy (in average number of bits per quantization level) is given by

$$H = - \sum_{j=1}^4 P\{x_j\} \log_2 P\{x_j\} = -2 p_1 \log_2 p_1 - 2 p_2 \log_2 p_2.$$

Verify that $H = 1.9015$ bits. [3 + 3]

Q. 4. [Error Probabilities: Coded versus Uncoded Scenarios]

Suppose uncoded PSK-modulated data is transmitted over an AWGN channel with $\frac{E_b}{N_0} = 9.6 \text{ dB}$. This data is now coded using (23, 12, 7) linear block code and transmitted over the same channel. Answer the following:

(i) Determine the number of bits the linear block code can detect and the number of bits it can correct. What is the code rate? **[2.5 Marks]**

(ii) Let E_c denote the average energy per coded bit. Compute $\frac{E_c}{N_0}$ (in dB). **[2 Marks]**

(iii) Determine the word error probabilities for the uncoded and coded communication systems. **[2 + 3 Marks]**

(iv) Suppose the requirement is that we need to achieve the codeword error probability computed in (iii) using the uncoded system by increasing the transmitted power, determine the required $\frac{E_b}{N_0}$ (in dB). **[3 Marks]**

Ans. (i) Given error control code: $(n, k, d_{min}) = (23, 12, 7) \Rightarrow n = 23, k = 12, d_{min} = 7$.

Error detection capability = $d_{min} - 1 = 6$.

Error correction capability: $t = 3$.

Code rate: $r = \frac{k}{n} = \frac{12}{23} = 0.5217$ [2.5]

(ii) $\frac{E_c}{N_0} = \frac{rE_b}{N_0} = 4.758 \Rightarrow \frac{E_c}{N_0} \text{ (dB)} = 6.77$ [2]

(iii) Word error probability (uncoded) $\approx k Q\left(\sqrt{2 \frac{E_b}{N_0}}\right) = 12 Q(\sqrt{18.24}) = 1.17 \times 10^{-4}$.

Word error probability (coded) $\approx {}^{23}_4C Q\left(\sqrt{2 r \frac{E_b}{N_0}}\right)^4 = 9.53 \times 10^{-9}$. [2 + 3]

(iv) We require $k Q\left(\sqrt{2 \left(\frac{E_b}{N_0}\right)_{eff}}\right) = 9.53 \times 10^{-9}$.

Solving, we get $\left(\frac{E_b}{N_0}\right)_{eff} = 18.21 \Rightarrow \left(\frac{E_b}{N_0}\right)_{eff} \text{ (dB)} \approx 12.6 \text{ dB}$ [3]

□□□□ **END** □□□□